Random Forest and Decision Tree

2026-04-24

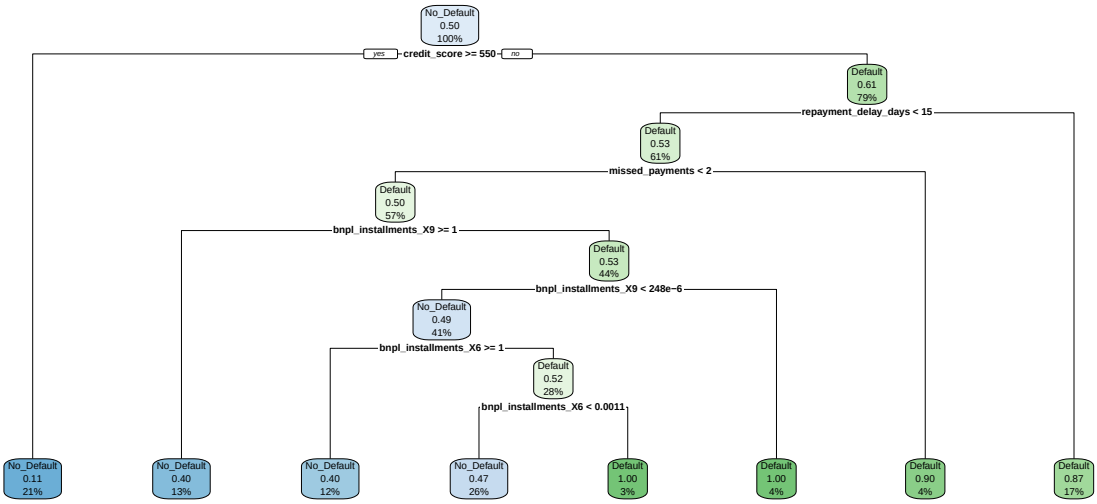
1. Model Development

Two models were developed to predict customer default risk: Decision Tree and Random Forest. These models were selected to compare interpretability and predictive performance.

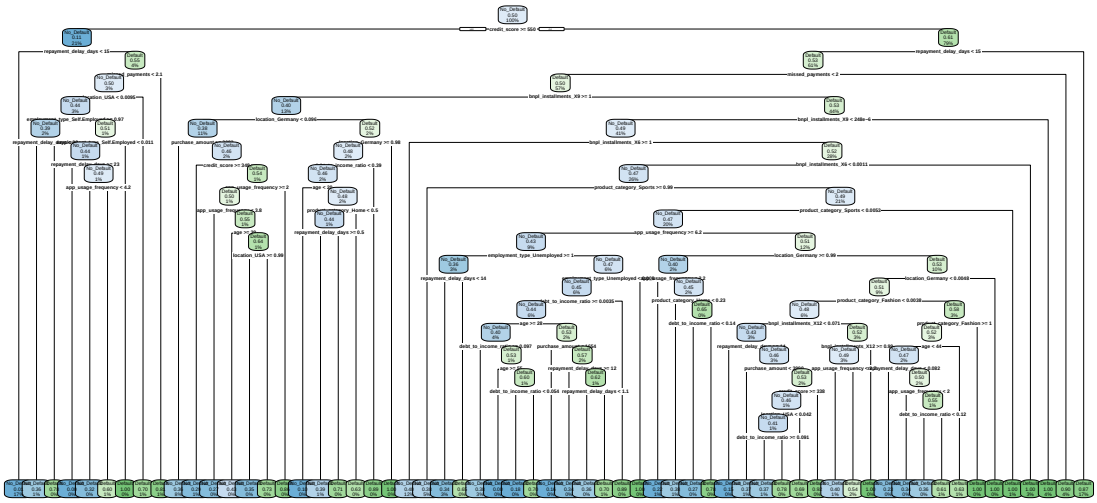
2. Decision Tree Model

2.1 Tree Visualization

Decision Tree (Default)



Decision Tree (Pruned)



2.2 Tree Complexity (Leaf Count)

Table 1: Decision Tree Complexity (Leaf Count)

Model	Leaves	Splits	Nodes
Decision Tree (Default)	8	7	15
Decision Tree (Full)	1570	1569	3139
Decision Tree (Pruned)	61	60	121

2.3 Validation Predictions (Sample)

Table 2: Decision Tree Predictions (First 10 Validation Rows)

Actual	DT_Default_Class	DT_Default_Prob	DT_Full_Class	DT_Full_Prob	DT_Pruned_Class	DT_Pruned_Prob
No_Default	No_Default	0.4738	No_Default	0	Default	0.6111
No_Default	Default	0.8715	Default	1	Default	0.8715
No_Default	No_Default	0.4738	No_Default	0	No_Default	0.3814
No_Default	No_Default	0.4738	No_Default	0	No_Default	0.3814
No_Default	Default	0.8715	Default	1	Default	0.8715
No_Default	No_Default	0.4031	No_Default	0	No_Default	0.4031
Default	No_Default	0.4031	Default	1	No_Default	0.4031
Default	No_Default	0.4029	No_Default	0	No_Default	0.3591
No_Default	No_Default	0.4738	Default	1	No_Default	0.2273
No_Default	No_Default	0.1146	No_Default	0	No_Default	0.3636

2.4 Performance

Decision tree performance on the validation set:

Table 3: Decision Tree Performance (Validation Set)

Model	Accuracy	Sensitivity	Specificity	Recall	Precision	F1	AUC
Decision Tree (Default)	0.717	0.351	0.945	0.351	0.801	0.488	0.738
Decision Tree (Full)	0.634	0.514	0.709	0.514	0.525	0.519	0.616
Decision Tree (Pruned)	0.702	0.471	0.848	0.471	0.659	0.549	0.751

Table 4: Confusion Matrix - Decision Tree (Default)

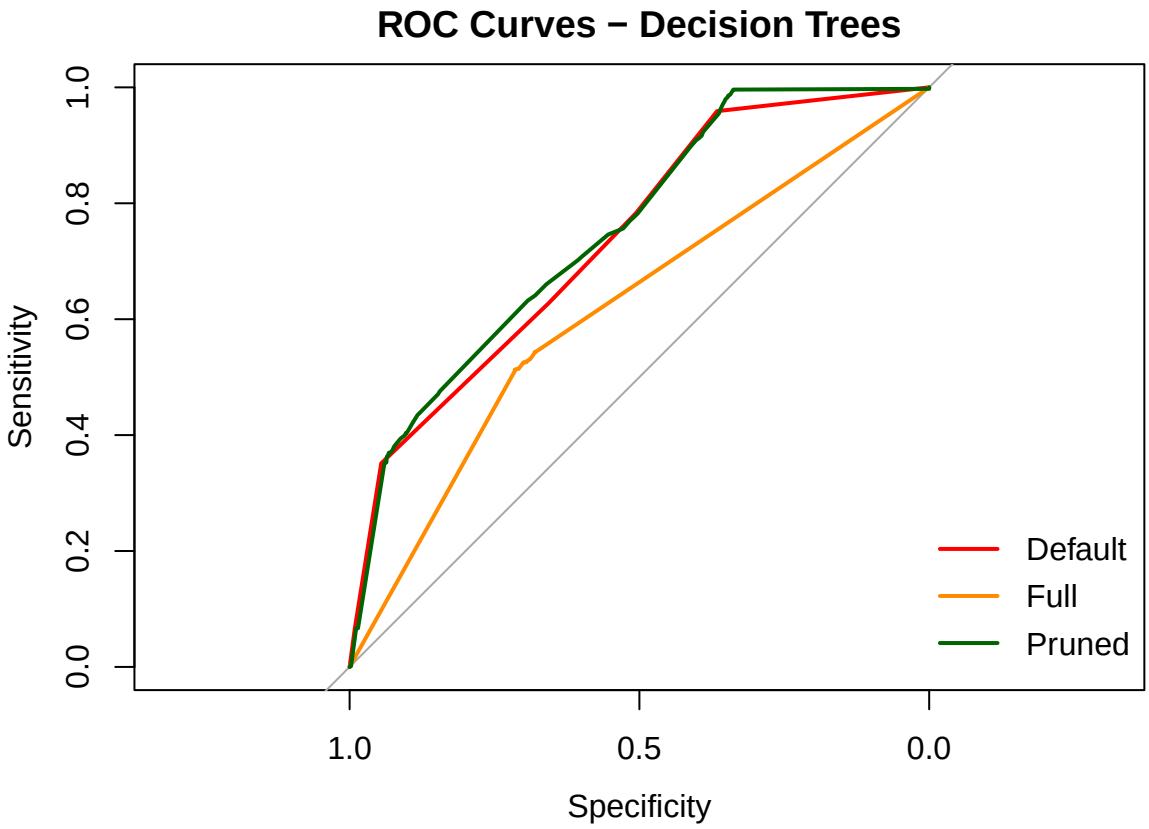
	No_Default	Default
No_Default	1178	506
Default	68	274

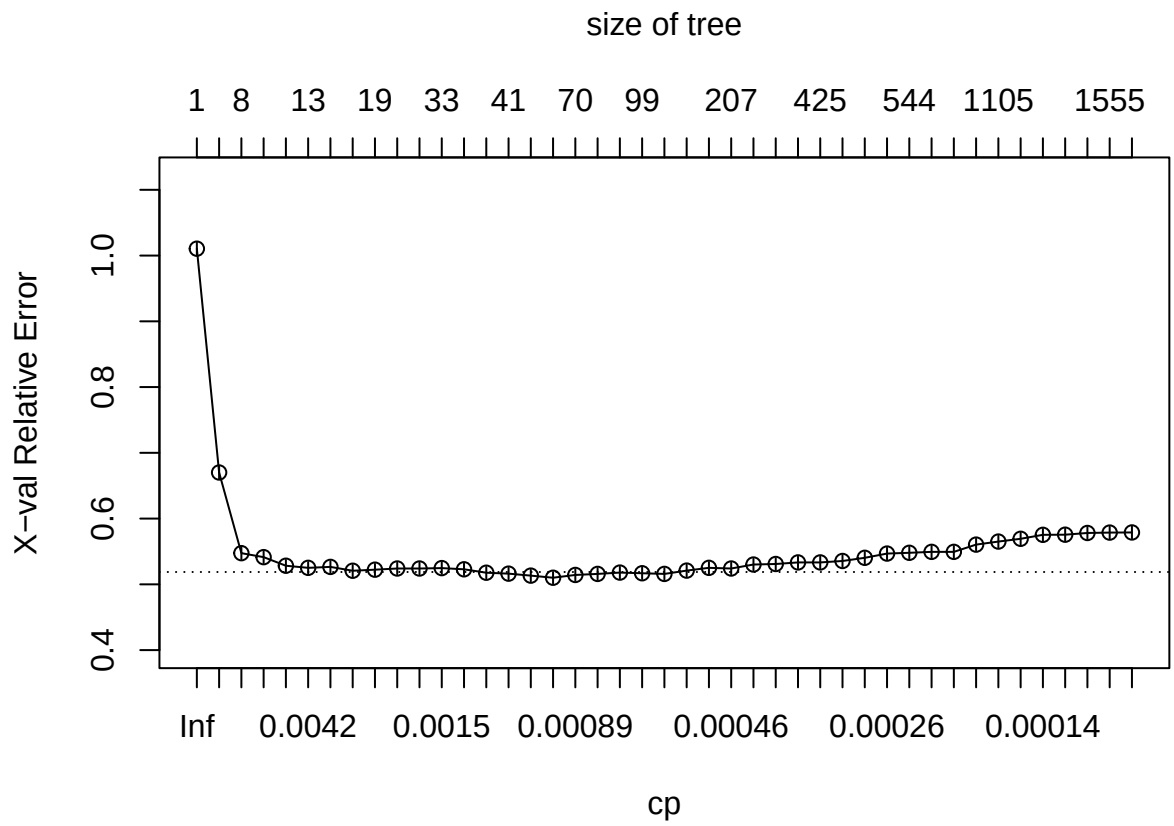
Table 5: Confusion Matrix - Decision Tree (Full)

	No_Default	Default
No_Default	883	379
Default	363	401

Table 6: Confusion Matrix - Decision Tree (Pruned)

	No_Default	Default
No_Default	1056	413
Default	190	367





3. Random Forest Model

```
## 0.02735043 0.01
## 0.009226714 0.01
## -0.04261863 0.01
```

3.1 Performance

Random forest performance on the validation set:

Table 7: Random Forest Performance (Validation Set)

Model	Accuracy	Sensitivity	Specificity	Recall	Precision	F1	AUC
Random Forest (Default)	0.700	0.447	0.859	0.447	0.665	0.535	0.761
Random Forest (Tuned mtry)	0.710	0.395	0.907	0.395	0.726	0.512	0.758
Random Forest (All features)	0.697	0.476	0.836	0.476	0.645	0.548	0.765

Table 8: Confusion Matrix - Random Forest (Default)

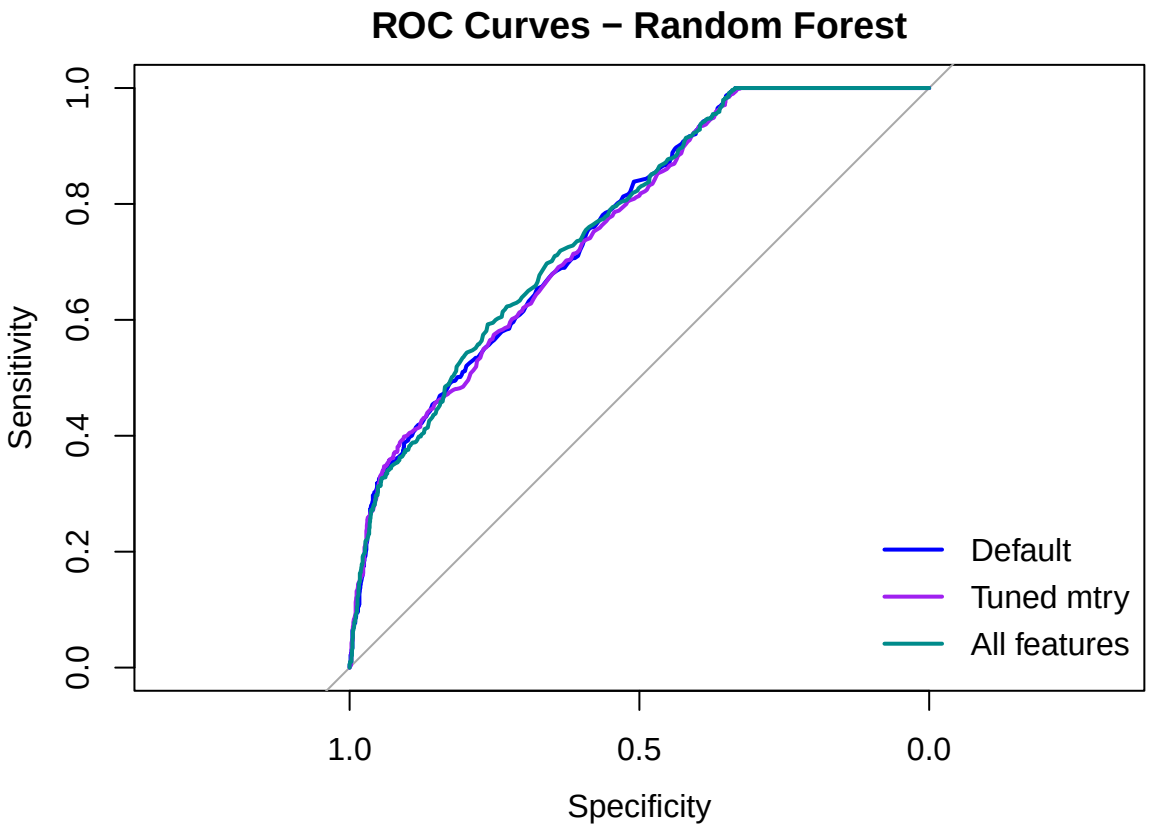
	No_Default	Default
No_Default	1070	431
Default	176	349

Table 9: Confusion Matrix - Random Forest (Tuned mtry)

	No_Default	Default
No_Default	1130	472
Default	116	308

Table 10: Confusion Matrix - Random Forest (All features)

	No_Default	Default
No_Default	1042	409
Default	204	371



3.2 Validation Predictions (Sample)

Table 11: Random Forest Predictions (First 10 Validation Rows)

Actual	RF_Default_Class	RF_Default_Prob	RF_Tuned_Class	RF_Tuned_Prob	RF_AllFeat_Class	RF_AllFeat_Prob
No_Default	No_Default	0.422	No_Default	0.482	No_Default	0.402
No_Default	Default	0.898	Default	0.894	Default	0.910
No_Default	No_Default	0.300	No_Default	0.252	No_Default	0.272
No_Default	No_Default	0.336	No_Default	0.272	No_Default	0.368
No_Default	Default	0.796	Default	0.788	Default	0.810
No_Default	No_Default	0.352	No_Default	0.274	No_Default	0.304
Default	No_Default	0.410	No_Default	0.404	Default	0.526
Default	No_Default	0.356	No_Default	0.294	No_Default	0.330
No_Default	No_Default	0.468	No_Default	0.422	No_Default	0.470
No_Default	No_Default	0.398	No_Default	0.332	No_Default	0.432

3.3 Feature Importance

The feature importance below is computed from the best random forest variant by validation AUC.

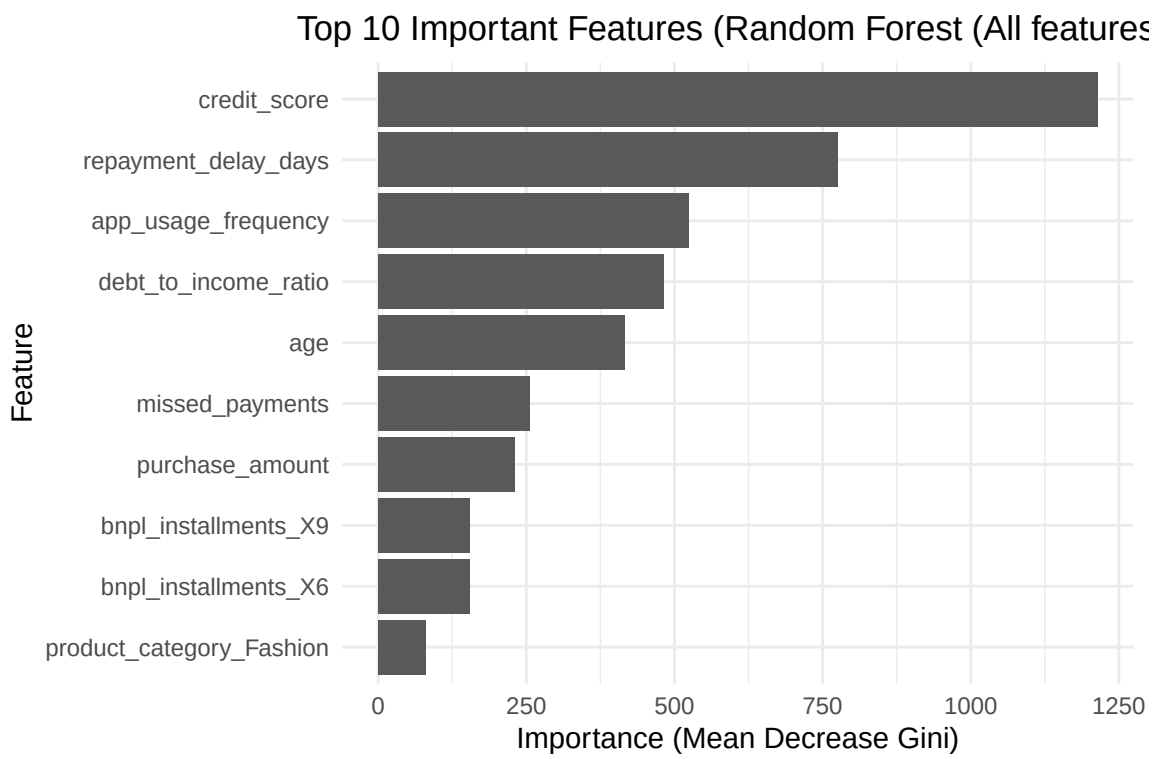


Table 12: Top 10 Important Features (Random Forest (All features))

Feature	MeanDecreaseGini
credit_score	1213.76
repayment_delay_days	775.96
app_usage_frequency	524.11

Feature	MeanDecreaseGini
debt_to_income_ratio	482.25
age	416.48
missed_payments	255.01
purchase_amount	230.79
bnpl_installments_X9	154.48
bnpl_installments_X6	154.20
product_category_Fashion	80.50

3.4 Performance Summary (Best Random Forest)

Table 13: Random Forest Performance

Metric	Value
Accuracy	0.697
Sensitivity	0.476
Specificity	0.836
AUC	0.765

4. Model Comparison

Table 14: Model Comparison (Validation Set)

Model	Accuracy	Sensitivity	Specificity	Recall	Precision	F1	AUC
Decision Tree (Default)	0.717	0.351	0.945	0.351	0.801	0.488	0.738
Decision Tree (Full)	0.634	0.514	0.709	0.514	0.525	0.519	0.616
Decision Tree (Pruned)	0.702	0.471	0.848	0.471	0.659	0.549	0.751
Random Forest (Default)	0.700	0.447	0.859	0.447	0.665	0.535	0.761
Random Forest (Tuned mtry)	0.710	0.395	0.907	0.395	0.726	0.512	0.758
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5. Interpretation

This section explains *why* each modeling step is included, and how to interpret the validation results.

5.1 How to read the metrics

We evaluate every model on the **validation set** (`validation_df`, currently set to `test_df`) using:

- **AUC**: overall ranking quality across all probability thresholds (higher is better).
- **Sensitivity (Recall)**: among actual defaulters, how many we correctly flag as default (higher reduces missed high-risk customers).

- **Specificity**: among non-defaulters, how many we correctly approve (higher reduces false alarms / unnecessary rejection).
- **Accuracy**: can be misleading if the classes are imbalanced, so we use it only as a supporting metric.

The best model by validation AUC is **Random Forest (All features)** (AUC = 0.765).

5.2 Decision Tree: why default → full → pruned

(1) Default tree (`dt_default`)
We start with default settings as a baseline: it is fast to train, easy to visualize, and gives an interpretable "first-pass" model. It also helps verify that the data and target are set up correctly.

(2) Full tree (`dt_full`, cp = 0)
We intentionally allow the tree to grow very large to observe the *maximum* complexity the training data can support. This is useful because it typically: - fits training patterns (including noise) very closely, and - exposes overfitting risk when validation performance does not improve.

In this run, the full tree has **1570** leaves (vs **8** for the default tree).

(3) Pruned tree (`dt_pruned`)
Pruning is used to trade a small amount of training fit for better generalization and simpler rules. Here we prune the full tree using the **complexity parameter** chosen by cross-validation: - selected `cp` $= 9.753 \times 10^{-4}$ (minimum cross-validated error in `dt_full$cptable`) - pruned tree leaves = **61**

If pruning improves or maintains validation AUC while using fewer leaves, it suggests the full tree was capturing noise rather than signal.

5.3 Random Forest: why default → tune `mtry` → all-features

(1) Default random forest (`rf_default`)
This is the baseline ensemble. Compared with a single tree, random forests reduce variance by averaging many decorrelated trees.

(2) Tuned `mtry` (`rf_tuned`)
`mtry` controls how many features are randomly considered at each split. Tuning it is a standard step because it changes the diversity/strength trade-off of trees in the ensemble.
We tune `mtry` using **out-of-bag (OOB) error** via `tuneRF()`, then refit with `ntree = 500`: - tuned `mtry` = **2**

(3) All features per split (`rf_all_features`)
Setting `mtry = p` (all predictors) is close to "bagging-style" trees. This is a useful comparison point: if this performs worse than tuned/default RF, it indicates that random feature selection is contributing to improved generalization.

5.4 Practical takeaways

- If your goal is to **catch as many defaulters as possible**, prioritize **Sensitivity** (and consider adjusting the decision threshold on predicted probabilities).
- If your goal is to **avoid rejecting good customers**, prioritize **Specificity**.
- Use the **pruned tree** when you need simple, explainable rules; use the **best RF** when you need stronger predictive performance.

6. Business Implications

The results suggest that customer default risk is influenced by complex interactions between multiple factors. The Random Forest model can be used to support risk-based decision-making, such as adjusting credit limits or implementing targeted risk controls.

By leveraging these insights, businesses can reduce financial losses while maintaining customer growth through more informed lending strategies.